# Huawei Access Node QoS Congestion Event Troubleshooting and Resolution SOP (YEL-ACC-52)

**Node:** YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2766

## Description

This SOP outlines the diagnostic and resolution procedures for a QoS congestion event on the Huawei Access Node YEL-ACC-52, located at the Yelahanka Sub-station. The congestion event is characterized by microbursts, interface tail drops, and QoS queue starvation, resulting in a Major alarm (ALARM_2766). The underlying cause of this failure domain is the interaction between the Huawei Access Node's hardware and software components, specifically the Quality of Service (QoS) module and the line card's buffering capabilities. The QoS module is responsible for prioritizing and policing traffic based on user-defined policies, while the line card's buffering capabilities can lead to packet drops and queue starvation when the traffic exceeds its capacity. When the QoS module is unable to manage the traffic effectively, it can result in microbursts and interface tail drops, ultimately leading to a QoS congestion event and the associated Major alarm.

## Pre-requisite Checks

- Verify that the Yelahanka Sub-station is accessible and that the remote hands are on site.

- Check the redundant link status to ensure that it is active and functioning correctly.

- Confirm that the Huawei Access Node YEL-ACC-52 is powered on and in a stable state.

- Verify that the necessary permissions and access rights are in place to execute the required commands.

## Potential Root Causes

| Cause | Probability |
|---|---|
| QoS policy misconfiguration | High |
| Insufficient line card buffering capabilities | Medium |
| Hardware failure on the Huawei Access Node | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display qos configuration, Expected Output: QoS configuration information, including policy names, priorities, and policing rates., Description: This command displays the QoS configuration information, allowing the technician to verify that the QoS policy is correctly configured and that the priorities and policing rates are set correctly.

Command: display interface statistics, Expected Output: Interface statistics, including packet drops, errors, and queue lengths.,
```

```
Description: This command displays the interface statistics, allowing the technician to identify any interface-related issues that may be contributing to the QoS congestion event.

Command: display qos policy statistics, Expected Output: QoS policy statistics, including policy hits, misses, and drops., Description: This command displays the QoS policy statistics, allowing the technician to verify that the QoS policy is functioning correctly and that there are no issues with policy hits, misses, or drops.
```

## Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: undo qos policy , Description: This command removes the specified QoS policy, allowing the technician to correct any misconfigurations and reapply the policy with the correct settings.

Step 2: Step: Adjust line card buffering capabilities, Command: set linecard buffering-capacity , Description: This command adjusts the line card's buffering capabilities, allowing the technician to increase the capacity and reduce the likelihood of packet drops and queue starvation.

Step 3: Step: Replace faulty hardware, Command: replace hardware , Description: This command replaces the faulty hardware component, allowing the technician to resolve the issue and prevent further QoS congestion events.

## Post-Check Verification

```
Command: display qos configuration, Expected Output: QoS configuration information, including policy names, priorities, and policing rates., Description: This command verifies that the QoS policy is correctly configured and that the priorities and policing rates are set correctly.

Command: display interface statistics, Expected Output: Interface statistics, including packet drops, errors, and queue lengths., Description: This command verifies that the interface-related issues have been resolved and that there are no further packet drops or queue starvation issues.

Command: display qos policy statistics, Expected Output: QoS policy statistics, including policy hits, misses, and drops., Description: This command verifies that the QoS policy is functioning correctly and that there are no issues with policy hits, misses, or drops.
```

## Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo qos policy , Description: This command restores the original QoS policy configuration, allowing the technician to revert to the previous settings and prevent any further issues.

- Step: Restore original line card buffering capabilities, Command: undo set linecard buffering-capacity , Description: This command restores the original line card buffering capabilities, allowing the technician to revert to the previous settings and prevent any further issues.

- Step: Restore original hardware configuration, Command: undo replace hardware , Description: This command restores the original hardware configuration, allowing the technician to revert to the previous settings and prevent any further issues.